Ashwini Khodre

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Mechanical Engineering
Indian Institute Of Technology, Jammu

GitHub | LinkedIn

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year of Passing
B.Tech (Mechanical Engineering)	Indian Institute of Technology, Jammu	7.68 (Till 7th sem)	2026
12th Standard	Queen Mary's HR. SEC. School	90%	2021
10th Standard	Queen Mary's HR. SEC. School	89.8%	2019

Experience

- **Tuning Bill** *Current*
  SDE Intern Jaipur
  - Developed a Banking System web application using ASP.NET Core (C) with secure user authentication, role-based access, and APIs for core banking operations.
  - Implemented account management and transaction workflows using Entity Framework Core and SQL Server, ensuring data integrity, validation, and clean backend architecture.

- **IIT Gandhinagar** *Jun. 2025 - Aug. 2025*
  Research Intern Gujarat
  - Devised and implemented a shell scripting pipeline that produced 500+ Abaqus input files from Neper-generated microstructures in Linux, which enabled comprehensive FEM simulations.
  - Constructed machine learning models, achieving 92% validation accuracy in predicting material behavior by leveraging geometric and statistical features derived from complex tessellation data.

Projects

- **Email Prioritizer** *Aug. 2025*
  AI-powered email management system GitHub
  - **Tools & technologies used**: Flask, Pandas, DBMS, Scikit-learn, Plotly, NLTK
  - Designed a full-stack web application featuring RESTful APIs and an interactive dashboard built with Plotly/Dash, reducing API latency by 20% through optimized data fetching and caching mechanisms.

- **Student Grievance Management System** *Dec. 2024*
  Student complaint portal GitHub
  - **Frameworks & Languages applied**: Django, JavaScript, Python, DBMS, HTML
  - Created a web-based platform for students to submit and track grievances while enabling administrators to manage and resolve issues efficiently. Applied a Third Normal Form (3NF) database structure to ensure data integrity, reduce redundancy, and streamline the resolution process.

- **Heart Disease Detection** *Jul. 2024*
  - **Platform & Tools used**: Excel, Numpy, Python, Matplotlib, Scikit Learn, Jupyter Notebook
  - Pioneered an innovative machine learning model identifying patterns in medical data from over 10,000 patient records, culminating in a precision/recall score of 0.9, enhancing risk stratification capabilities.

Technical Skills

- **Programming & Scripting:** C++, C #, GO , C, Python, SQL, Shell Scripting
- **Data Science & ML:** Pandas, NumPy, Scikit-Learn, Matplotlib, TensorFlow, Jupyter Notebook, Google Colab
- **Software & Tools:** Visual Studio, Postman, SQL Server, Excel
- **Frameworks & Development:** ASP.NET Core, Django, Data Preprocessing, Feature Engineering
- **Operating Systems:** Linux, Windows

Key Courses Taken

- Computer Networks, DBMS, Data Structures and Algorithms, Numerical Method, Thermal Energy System, Fluid Dynamics, Heat Transfer, Linear Algebra, Calculus, CAD.

Achievements

- **Solved 300+ Data Structures and Algorithms problems on LeetCode**
- **Amongst top 10 students** in Mechanical Engineering in IIT Jammu. *2024*
- **Ranked 3rd in hand made RC car competition** Won price worth Rs 6000. *2023*

Positions of Responsibility

- **Journalism Head** Prism, Media Cell-Coordinated 12+Members, Published 15+articles *Jun. 2024 - Apr. 2025*
- **Event Management head** Astronomy Club-Organized 5+ events for 200+participants *2023-24*
- **Infra Co-Head ANHAD** Managed 10+volunteers, coordinated 3 stage setups for 1000+attendees. *Mar. 2024*

CO-CURRICULAR ACTIVITIES

- Captain, IIT Jammu Basketball Team — Represented in Inter IIT Sports Meet.
- 2 times Gold Medalist in Intra-IIT Basketball Tournament.
- Secured 3rd position in Dance Competition.